

EC 504

Final Exam Practice, Fall 2025

Note: This is much longer than the final exam will be, but it has questions covering all of the possible topics in the exam. Use it for practice. The real exam will be shorter.

Problem 1

Answer True or False to each of the questions below, **AND YOU MUST PROVIDE SOME FORM OF EXPLANATION (very brief is OK)**. Each question is worth 2 points. Answer true only if it is true as stated, with no additional assumptions.

- (a) In a forward star representation of a directed graph, the edges must be sorted by start node.
- (b) Suppose T_1, T_2 are minimum spanning trees for the same graph G . Then the largest edge weight of T_1 must be the same weight as the largest edge weight of T_2 .
- (c) In Bellman-Ford's shortest path algorithm, the temporary distance labels D assigned to vertices which have not yet been scanned (i.e. left the work queue) can be interpreted as the minimum distance among all paths from the start vertex to the given vertices with intermediate vertices restricted to the vertices that have already been scanned (left the work queue).
- (d) Consider a minimum spanning tree T in a weighted, undirected graph G . Suppose we decrease the weight of a single edge that is in the minimum spanning tree T . Then, it is possible that T is no longer a minimum spanning tree in the modified graph.
- (e) Consider a minimum spanning tree problem in a connected, weighted graph with positive and negative weights which can include negative weight cycles. Then, both Prim's and Kruskal's algorithms can be used to find the minimum weight spanning tree in this graph.
- (f) In the KMP algorithm for string matching, the prefix function for the string "AAABBBBA" is $[0,1,2,0,1,2,1]$.
- (g) Consider a minimum spanning tree T in a weighted, undirected graph G . Suppose we decrease the weight of a single edge **that is in the minimum spanning tree** T . Then, it is possible that T is no longer a minimum spanning tree in the modified graph.
- (h) A problem is said to be NP-complete if it can be reduced in polynomial time to an instance of every other problem which is in class NP.
- (i) Given a weighted, directed, acyclic graph with non-negative weights, consider the problem of finding the longest path from node 1 to every other node. We can solve this problem using the Bellman-Ford algorithm.
- (j) Consider a minimum spanning tree T in a connected, weighted undirected graph (V, E) where the weights have different values. Then, for every node i , the minimum spanning tree must contain the arc $\{i, j\}$ with the smallest weight connected to node i .
- (k) Dynamic programming can be used to solve the integer knapsack problem in pseudo-polynomial time.
- (l) The sequence alignment problem can be solved using dynamic programming in time polynomial in the lengths of the two sequences to be aligned.
- (m) Consider a 2-d binary search tree that was constructed by branching on the median value at each level, storing the keys at the leaves of the tree. Assume there are n keys in the tree, where each key is an ordered pair of values. The worst case complexity to find a key is $O(\log(n))$.
- (n) Consider an undirected graph where, for every pair of nodes i, j , there exists a unique simple path between them. This graph must be a tree.
- (o) Prim and Kruskal's algorithm can also be used to find the maximum weight spanning tree in a graph.
- (p) Consider a connected, undirected graph. Then, the worst case computational complexity of the Floyd-Warshall algorithm for finding the shortest distance for every pair of nodes does not depend on the number of edges in the graph.

Problem 2

Consider the following 2-dimensional set of points:

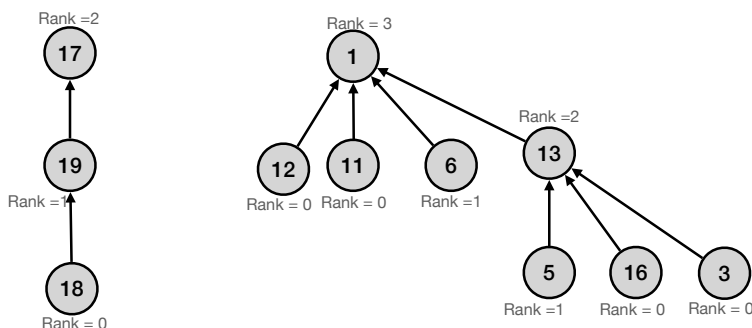
$$(80, 90), (85, 78), (60, 40), (10, 10), (15, 15), (30, 80)$$

- Arrange the elements in a PR quadtree with value range 0-96 for both the first and second coordinates. Display the answer **as a tree**, with children arranged left-to-right in the following order: SW, SE, NW, NE.
- Form a balanced 2-d tree using the median of the elements to split, with all the keys at the leaves of the tree. When there are an even set of points to split at vertex, select the smaller of the two possible median values as the navigation key.

Problem 3 Draw the Aho-Corasick trie corresponding to the following search patterns: SPEAK, LEAK, EAT, PEAR, APPEAR. Show the suffix links, except for the suffix links that revert back to the root. Don't bother showing the output links.

Problem 4

Consider the two trees, shown on the right, that are part of a disjoint set forest. Suppose we find a relation between keys 16 and 18. Show the disjoint set tree that results from the union operation using merge by rank and path compression, where, if two roots have the same rank, merge the larger value root under the smaller value root.



Problem 5

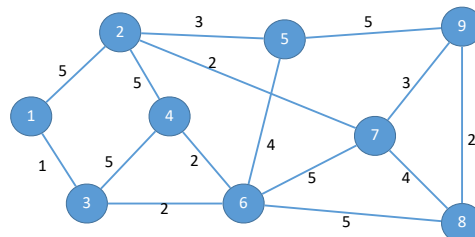
Consider the two words: *people* and *purple*. Consider them as a sequence of letters. We would like to find the best alignment of the two letter sequences, with the following costs:

- Incur a penalty of 1 for any skipped letter.
- Incur a penalty of 3 for any mismatched letter .
- Incur a reward of 2 for correctly matching two letters.

Using this reward structure, find the minimum cost alignment. State the cost of the alignment.

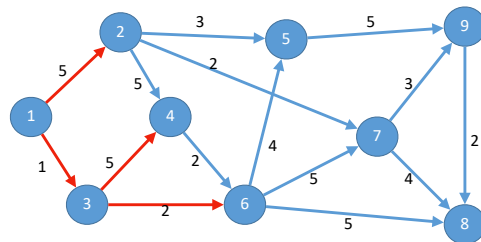
Problem 6

Consider the undirected graph on the right, where the numbers on edges consist of weights. Find a minimum spanning tree for the graph on the right. Indicate what algorithm you are using, and show some of the work associated with that algorithm. Display the minimum spanning tree graphically, and show the total weight of the minimum spanning tree.

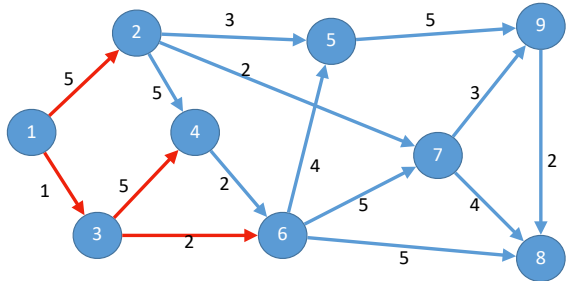
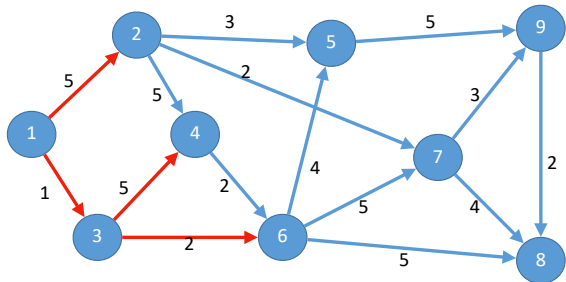
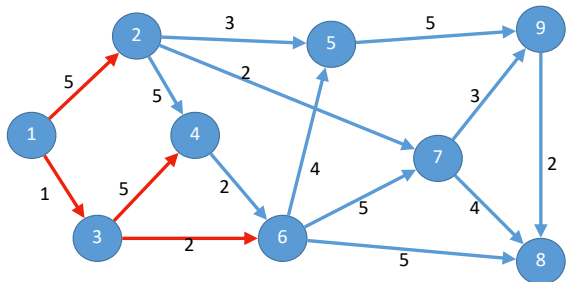
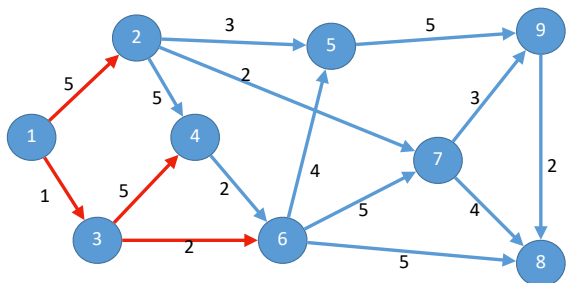


Problem 7

Consider the directed graph on the right, where the numbers on edges consist of distances. We are going to find a shortest path tree from node 1 to every other node using Dijkstra's algorithm. The figure shows the partial tree grown after five iterations, scanning vertex 1, then vertex 3, then vertex 6, then vertex 2, and then vertex 4.

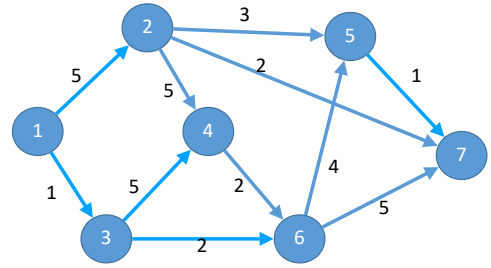


- (a) Show the sequence of four paths that Dijkstra's Algorithm will find to solve the problem, by drawing the edges on the graphs below.



Problem 8

Consider the directed graph on the right, where the numbers on edges consist of edge capacities. Our goal is to find the max flow from vertex 1 to vertex 7. As such, we will do it using both the Ford-Fulkerson algorithm, using the Edmonds-Karp variation where shortest augmenting paths are found using breadth first search in the residual graph.



- Assume that we have made an augmentation of one unit of flow on path 1 - 3 - 6 - 7, and another augmentation of 2 units of flow on path 1 - 2 - 7. Construct the residual network, with residual capacities on each forward and reverse edge.
- On that residual network, find the shortest (least number of hops) augmenting path from vertex 1 to vertex 7, and determine its capacity.

Problem 9

Consider the following knapsack problem: We have six objects with different values and V_i and sizes w_i . The object values and sizes are listed in the table below:

Object number	1	2	3	4	5	6
Value	10	11	12	13	14	15
Size	1	1	2	3	4	3

- Suppose we are given a knapsack with total size 11, and you are allowed to use fractional assignments. What is the maximum value that fits in the knapsack for the above tasks?
- Assume we are not allowed fractional assignments. For the knapsack with total size 11, the dynamic programming requires computing $Opt(j, k)$, the best value considering only objects $1, \dots, j$, using total size k . The table below has partially computed these numbers. Compute the remaining elements in the table, $Opt(j, k), j = 5, 6; k = 5, 6, 7, 8, 9, 10, 11$.

Capacity-> Max .Task	0	1	2	3	4	5	6	7	8	9	10	11
0	0	0	0	0	0	0	0	0	0	0	0	0
1	0	10	10	10	10	10	10	10	10	10	10	10
2	0	11	21	21	21	21	21	21	21	21	21	21
3	0	11	21	23	33	33	33	33	33	33	33	33
4	0	11	21	23	33	34	36	46	46	46	46	46
5	0	11	21	23	33	34	36					
6	0	11	21	23	33	36	38					

- What is the optimal set of entire tasks that fit in the knapsack with size 11 and what is the maximum achieved value?